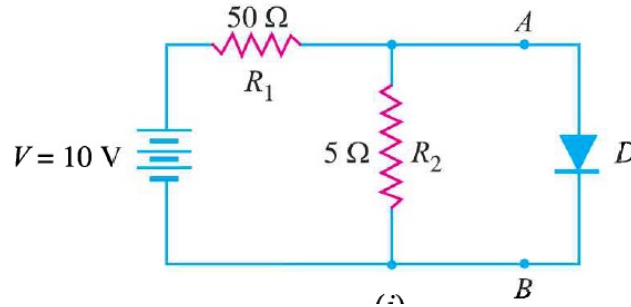


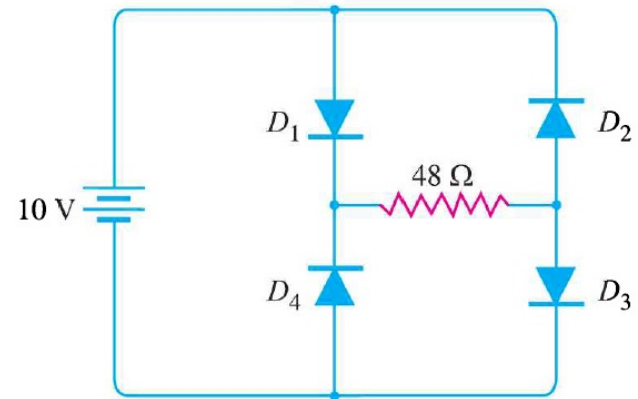
Unit 2 Supplementary Questions

- ▷ A *pn* junction is doped with $N_A = 2 \times 10^{16} \text{ cm}^{-3}$ and $N_D = 9 \times 10^{15} \text{ cm}^{-3}$. Determine the capacitance of the device with (a) $V_R = 0$ and $V_R = 1 \text{ V}$.
- ▷ A *pn* junction with $N_D = 3 \times 10^{16} \text{ cm}^{-3}$ and $N_A = 2 \times 10^{15} \text{ cm}^{-3}$ experiences a reverse bias voltage of 1.6 V . (a) Determine the junction capacitance per unit area. (b) By what factor should N_A be increased to double the junction capacitance?
- ▷ Due to a manufacturing error, the p-side of a *pn* junction has not been doped. If $N_D = 3 \times 10^{16} \text{ cm}^{-3}$, calculate the built-in potential at $T = 300 \text{ K}$.
- ▷ A junction employs $N_D = 5 \times 10^{17} \text{ cm}^{-3}$ and $N_A = 4 \times 10^{16} \text{ cm}^{-3}$. (a) Determine the majority and minority carrier concentrations on both sides. (b) Calculate the built-in potential at $T = 250 \text{ K}$, 300 K , and 350 K .

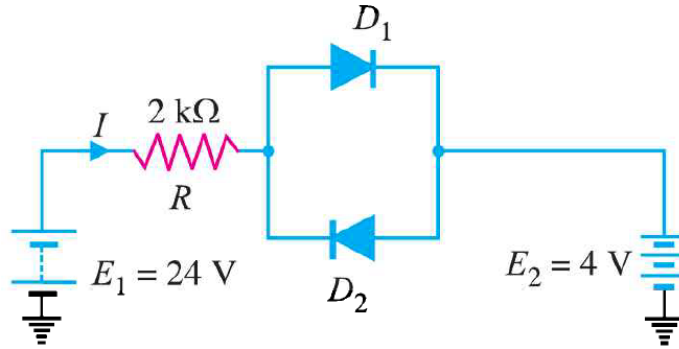
- Find the current through the diode in the circuit shown below. Assume the diode to be ideal



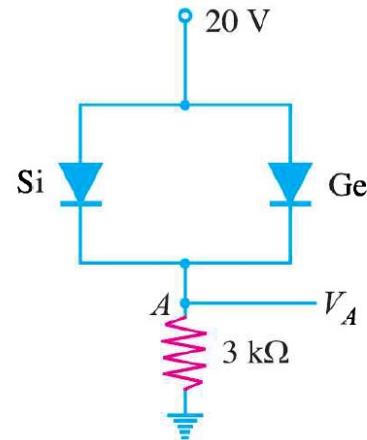
- Calculate the current through $48\ \Omega$ resistor in the circuit below. Assume the diodes to be of silicon and forward resistance of each diode is $1\ \Omega$.



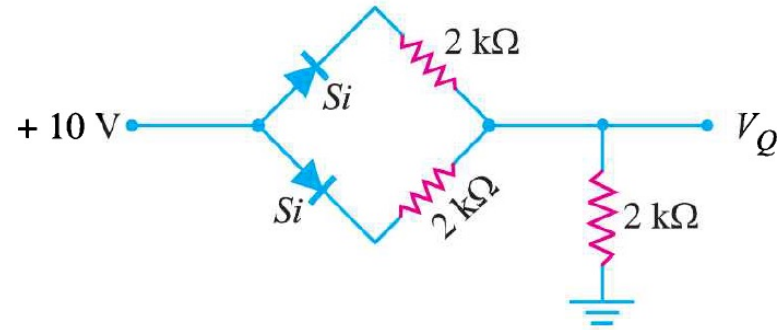
- Determine the current I in the circuit shown below. Assume the diodes to be of silicon and forward resistance of diodes to be zero.



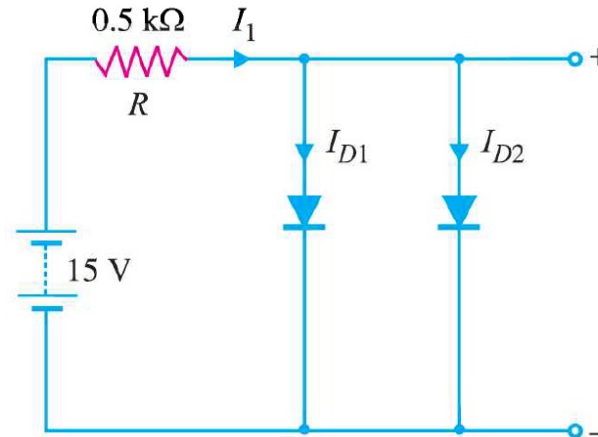
- Find the voltage V_A in the circuit shown below



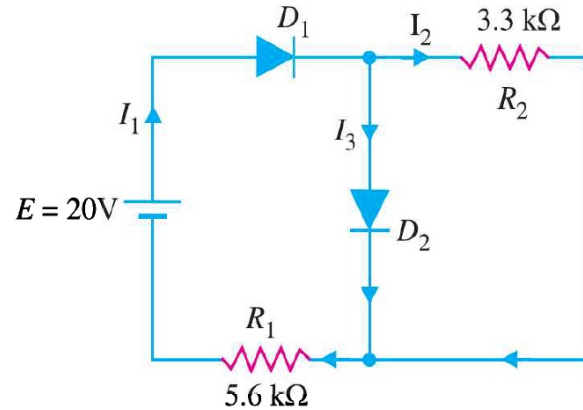
- Find V_Q and I_D in the network shown below



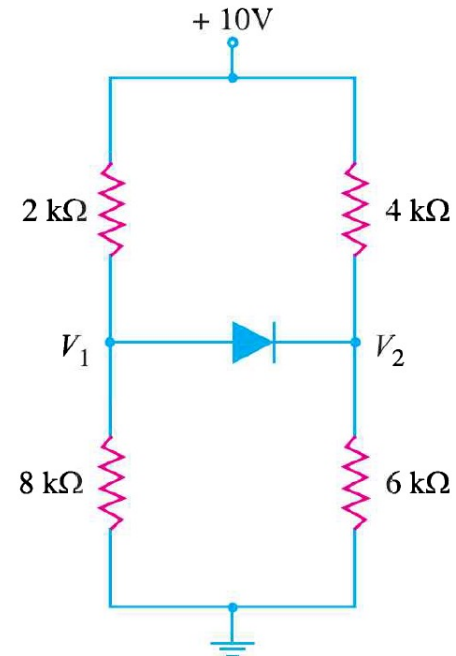
- Determine current through each diode in the circuit shown below. Assume diodes to be similar.



- Determine the currents I_1 , I_2 , and I_3 for the network shown below. Use simplified model for the diodes.



- Determine if the ideal diode in the figure to the right is forward bias or reverse bias.



Thanks!

Any questions?